

# DevOps to AI/ML Roadmap

## Phase 1: Master Conventional DevOps (Foundation)

- Version Control: Git, GitHub/GitLab
- CI/CD Pipelines: Jenkins, GitHub Actions, GitLab CI
- Build & Test Automation: Maven/Gradle, JUnit, Selenium
- Artifact Repositories: Nexus, JFrog Artifactory
- IaC Tools: Terraform, Ansible
- Containerization: Docker
- Orchestration: Kubernetes
- Cloud Providers: AWS / Azure / GCP
- Monitoring: Prometheus + Grafana
- Logging: ELK Stack (Elasticsearch, Logstash, Kibana), Fluentd

## Phase 2: Automate Everything (Efficiency & Consistency)

- Infrastructure Automation: Use Terraform & Ansible in pipelines
- Secrets Management: Vault, AWS Secrets Manager
- Policy as Code: OPA, Sentinel
- SLOs & SLIs
- Alerting: Alertmanager, PagerDuty, Opsgenie

## Phase 3: Transition to Intelligent DevOps (AI/ML Enablers)

- Learn Python for ML (pandas, NumPy)
- Intro to Machine Learning (scikit-learn, TensorFlow/Keras)
- Exploratory Data Analysis (EDA)
- Time series forecasting (Prophet, ARIMA)
- Log anomaly detection using ML
- Resource prediction
- Auto-scaling recommendations

## Phase 4: MLOps & AI-Augmented Pipelines

- Model Tracking & Versioning: MLflow, DVC
- Model Deployment: Docker, FastAPI, Kubernetes

- Model Monitoring: Prometheus/Grafana + custom metrics
- Add ML test validation into CI
- Trigger auto-retraining or model rollback based on metrics

## **Phase 5: AIOps - Full Intelligence in Operations**

- Incident Prediction & RCA: BigPanda, Moogsoft, Splunk AI
- Event Correlation & Noise Reduction: Dynatrace, Datadog AI
- Smart Alerting
- Self-healing Systems
- ChatOps with LLMs (e.g., GPT-based bots)

## **Phase 6: Custom AI DevOps Solutions (Advanced)**

- Train custom models on release failure patterns
- Classify change requests by risk level
- Auto-tune infrastructure (e.g., smart Kubernetes pod sizing)
- Use LLMs to auto-generate scripts/configs, explain logs, and document pipelines

## **Summary Flow (TL;DR)**

- DevOps Basics -> Automation & CI/CD -> Monitoring -> Learn ML -> AI for Logs & Metrics -> MLOps -> AIOps -> Custom AI Tools